

Reddy believes that qualification is required just to enter the industry. Catering to designs required by the industry actually depends on the design skills of the engineer, adaptability of technologies and new trends emerging in the field.

More than the educational qualification, it is the interest and the opportunity that make someone an analogue designer, points out Binu Raj S.

Entry-level roles

Dr Karthik informs that entry-level roles mainly include design and verification of analogue blocks. "Their designation would be that of a design engineer focusing on areas like circuit design, layout design and circuit characterisation," says Somashekhar.

Rajani Rao says, "Candidates would have had theoretical exposure on all these basic blocks in their academic courses. It is now time to actually design them and build on their fundamentals. They would also get an idea of how these building blocks get stitched into a complex analogue top level like a PLL."

According to Garadi, fresh graduates from universities would be testing or debugging boards under the tutelage of senior design engineers.

Analogue electronics is a field where an engineer's career grows slowly when dedicated to analogue sector alone, believes Reddy. Engineers need to start off from production or testing rather than designing in the entry level.

Reddy says, "Engineers need to concentrate on analogue assisted products like instrumentation, power or digital, so that they can have faster growth. The Indian industry requires engineers of that kind," he adds.

Pay package

"On an average, the starting salary of an application engineer in linear ICs is in the range of ₹ 600,000 per annum in some of the leading semiconductor companies. For an IC designer it can be about 30 to 50 per cent higher," informs Garadi.

Binu Raj believes that there is not much difference between the pay packages of an analogue designer and

Career in analogue design: Myth versus reality

Myth—It is a digital world and there is not much scope for analogue design.

Reality—Analogue semiconductor market is around \$40 billion and has been increasing steadily. The real-world signals—for example, touch, sound, gesture, video, pressure and temperature—are all analogue in nature. Digital processing techniques are used to manipulate the real-world data. However, analogue processing is used to capture and condition the input data as well as for output processing. As digital processing has advanced significantly over the last few decades, demands on analogue processing have also increased. This has resulted in the increased number of bits for data converters (ADCs and DACs) as well as the ever-increasing sample rate. High-speed wired interfaces like HDMI and PCIe, as well as wireless communication (RF) have increased the need and scope for analogue design significantly.

Myth—Analogue design is only about designing transistor-based circuits.

Reality—Analogue design is an inter-disciplinary science that requires knowledge of diverse fields like control systems, signal processing, network synthesis and electromagnetics. For example, modern data converter designs use signal processing techniques to achieve high resolution.

—Dr S. Karthik, engineering director, IPDC, Analog Devices India

a digital designer. "The only difference will be in the area of work," he says.

Somashekhar informs that the annual package for an entry-level B.Tech candidate would be between ₹ 350,000 and ₹ 400,000 per annum, whereas for an M.Tech graduate it would be around ₹ 400,000 to ₹ 500,000 per annum.

According to Reddy, pay package in the Indian electronics industry is a highly indefinite figure as it totally depends on the product that the industry is concentrating on.

Skills to be developed

Garadi believes that effective analogue circuit designs require a strong understanding of core linear devices and how they affect analogue circuit design. He says, "Analogue designs are a combination of discrete devices (BJTs, FETs, etc), ICs (more often than not, op-amps) and passives. Therefore a good understanding of discrete devices' behaviour in linear mode will hold one in good stead for analogue designs."

Garadi adds, "Having acquired the knowledge of discrete devices, op-amps would pose very little challenge in getting to know them. Be sure of which passive components you will use for what and be aware of their specific behaviour and functionalities." Component modeling and circuit simulation is yet another skill that could unleash the full potential of an analogue designer.

Reddy too feels that analogue basics and knowledge of components

and their behaviour must be strong. Patience in testing an analogue circuit is a skill that needs to be practised.

Inquisitiveness at a young age to experiment with analogue circuits plays an important role. Binu Raj says, "It could be at least the eagerness to open a radio or make some hobby circuit and feel the happiness when it works. That type of exercise makes one's analogue fundamentals strong."

The analogous conclusion

Garadi believes that although opportunities for analogue designers are numerous, there are fewer engineers wanting to take up analogue circuit design as a career. "On that count, if you have proved your mettle in analogue design, you would most certainly find a job that interests you and you will definitely do well in mixed-signal and digital design, where the opportunities are a lot more," he adds.

According to Somashekhar, until recently most of the design services outsourced to India were in the areas of software and digital VLSI design. Due to significant increase in analogue/mixed-signal content of the current/future-generation system-on-chips, there is huge demand for analogue engineers to support this design activity. There is acute shortage of analogue engineers globally and hence more of analogue design work is being outsourced to India. To support these increased design opportunities, there is need to create more analogue engineers. ●

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